

# The Graph Forces Principle

Casey Koons (named principle) + Grace (operational framing) + Elie (two-cluster-types extension)

2026-05-20 Wednesday EOD

## The Graph Forces Principle

Casey-named statement (operational register, per Cal Calibration #13)

“Substrate-algebraic patterns appear to emerge from the AC graph’s overdetermined-identity clustering, not from speculation. When  $\geq 4$  EXACT identities cluster around a common BST-primary form, the cluster FUNCTIONS AS a candidate substrate signature in the framework’s diagnostic. The ‘no accidents’ hypothesis is operationalized as graph diagnostic: overdetermined identities are treated as substrate-structure evidence pending falsification.”

This is an OPERATIONAL DIAGNOSTIC, not a metaphysical claim. The “functions as” register is load-bearing: BST does not claim that overdetermined identities ARE substrate signatures by ontological necessity; it operationally TREATS them as candidate evidence in the audit chain.

### Two cluster types in the diagnostic (Elie discovery)

Per Elie Toy 3152 + 3153 + 3158, three distinct cluster shapes function as substrate signatures:

Type 1 — Overdetermined-form cluster: - ONE NUMBER with multiple BST-primary algebraic forms - Example: Universal  $Q=126$  with 5 equivalent forms ( $M_g-1$ ,  $2^g\text{-rank}$ ,  $N_{\text{max}-c_2}$ ,  $n_{\text{Cl-derived}}$ ,  $N_{c\cdot C_2\cdot g}$ ) - Substrate signature: the same value reached via independent algebraic paths through BST primaries

Type 2 — Cross-domain anchor cluster: - ONE BST-PRIMARY NUMBER appearing in multiple independent physical domains - Example:  $\chi=24$  in K3 Euler char +  $SU(5)$  dim + heat kernel cascade + Niemeier rank + Witten-Reshetikhin-Turaev ( $\geq 5$  domains) - Substrate signature: same integer manifesting across structurally-independent contexts

Type 3 — Compound cluster (K72 candidate): - ONE NUMBER exhibiting BOTH Type 1 AND Type 2 simultaneously - Example:  $\chi=24$  has BOTH 5 Type-1 forms

AND  $\geq 6$  Type-2 domains = 30-fold multiplicative overdetermination - Strongest substrate signature in the diagnostic catalog

All three types function as substrate-signature candidates pending audit-chain verification (P1-P7 strict counting per K58 protocol).

### Operational falsifier (Grace Task #215 batch-test method)

The principle is operationally falsified by: - Density of overdetermined-EXACT-identity clusters  $\leq$  random-ring null model (K44 methodology extended) - If statistical scan of clusters across the graph shows no significant excess over null, the “no accidents” diagnostic fails - Current state (Toy 3138): 8/8 PASS with  $1.19\times$  null signal — modest excess; Mode 1 forward-prediction follow-up needed for closure

Specific falsifier: a future EXACT-identity cluster that, under audit, fails P1-P7 strict counting AND turns out to be statistical coincidence.

### What the framework does NOT claim

- NOT a proof that overdetermined identities ARE substrate signatures by metaphysical necessity
- NOT a closure on the “no accidents” question: the diagnostic is operational; the metaphysical question remains open
- NOT a claim that all BST observables exhibit overdetermined-clustering: many observables have single-form derivations and that’s structurally fine

### Tier discipline (per Cal Substrate-Cognition External Register methodology)

Level 1 (D-tier supportable internally): - The 6 EXACT identities catalogued at  $1e-14$  algebraic precision exist (Universal  $Q=126$ ,  $c_{FK}\cdot\pi^{(9/2)}=225$ , Tsirelson<sup>2</sup>- $S_{BST}^2=1/2^{N_c}$ , etc.) - These admit multiple BST-primary forms (Type 1 / Type 2 / Type 3 cluster shapes) - The catalog is reproducible

Level 2 (I-tier working diagnostic with falsifier): - Overdetermined-identity clustering FUNCTIONS AS substrate signature - “No accidents” diagnostic operationally validates BST-substrate framing - Mode 1 forward-prediction discipline pre-registers tests before measurement

### External register discipline

Internal use: Graph Forces Principle as operational diagnostic for substrate signature detection.

External use: ACCEPTABLE (Cal grade-pass not strictly required for operational claim): - “BST framework treats overdetermined-EXACT-identity clusters as candidate substrate evidence at I-tier per the no-accidents diagnostic” - “The cluster

identification protocol applies P1-P7 strict counting per K58” - Specific cluster examples ( $Q=126$ ,  $\chi=24$ ,  $c_{FK}=225$ ) acceptable with full algebraic-identity verification disclosure

Unacceptable external register: - “BST has proven that overdetermined identities are substrate signatures” - “The no-accidents principle is established” - Any unfalsifier-bearing form of the claim

### Connection to other principles

- SWPP: SWPP describes substrate operations; Graph Forces describes how we DETECT substrate signatures
- Integer Web Principle: webs intersect at overdetermined-identity points (cross-web crossings = Type 1/2/3 clusters)
- M2C2 (Multi-CI Convergent Calibration): M2C2 is the process-level analog of Graph Forces at claim-level (multi-CI convergence is multi-channel substrate signature)
- K-audit chain: K-audit candidates K54 (3/1507 family), K61 (131 family), K69 (Universal  $Q=126$ ), K72 (compound cluster  $\chi=24$ ) are all Graph Forces diagnostic outputs

### Per Casey’s standard

- Simple: count overdetermined-identity clusters; compare to null model
- Works: Grace Task #215 batch-test produced  $1.19\times$  null signal (modest above-noise)
- Hard to break: P1-P7 strict counting per K58 + Mode 1 forward-prediction discipline
- Counter-example: future positive Mode 1 failure would invalidate diagnostic; not currently observed

### Standing applications

Active K-audit candidates that function as Graph Forces diagnostic outputs: - K54 BST fine-structure family at 3/1507 (5 contexts) - K61 Family  $Q=131$  (3 strong + 1 conditional) - K69 Universal  $Q=126$  (5 BST-primary forms) - K72 candidate (compound cluster  $\chi=24$  with 30-fold multiplicative overdetermination) - Integer-10 over-determination candidate

Each functions as substrate-signature evidence operationally; audit chain confirms or refines.

### Filing status

Casey-named principle FILED per Wednesday 2026-05-20 approve-all directive. L2 operational diagnostic with explicit falsifier (Grace Task #215 batch-test method). Internal and external register acceptable per disclosed framing.

— Keeper, 2026-05-20 Wednesday EOD (Cal register discipline applied)